

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: An Observational Study of Effectiveness and Safety of the First-Line Nivolumab Plus Ipilimumab With or Without Chemotherapy for Advanced/Recurrent Non-Small Cell Lung Cancer in Japan **Short Title/Acronym:** LIGHT-NING Study **Protocol Number:** CA209-64A / UMIN000044375 **NCT Number:** NCT05161325 **Sponsor Name:** Bristol-Myers Squibb / Ono Pharmaceutical Co., Ltd. **Investigational Product:** Nivolumab (Opdivo) plus Ipilimumab (Yervoy) with or without chemotherapy **Phase of Development:** Observational / Real-world Data (N/A for Phase) **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the effectiveness and safety of Nivolumab plus Ipilimumab with or without chemotherapy as first-line treatment for participants with untreated advanced or recurrent non-small cell lung cancer (NSCLC) in the real-world setting in Japan. **Secondary Objectives:** To evaluate progression-free survival (PFS), overall survival (OS), the incidence of grade 3-4 immune-related adverse events (irAEs), and overall treatment continuation rates in standard clinical practice.

2.2 Background & Rationale Historically, platinum-based doublet chemotherapy has been the standard first-line treatment for NSCLC. While recent phase III trials (such as CheckMate 227 and CheckMate 9LA) demonstrated the survival benefit of nivolumab plus ipilimumab with or without chemotherapy, their strict inclusion criteria limited their external validity. This observational study was designed to generate real-world evidence confirming the safety, effectiveness, and treatment patterns of this regimen in a broader, everyday clinical population in Japan.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Multicenter, non-interventional, retrospective cohort study) **Control Type:** Single-arm (Evaluated in two sub-cohorts: with or without chemotherapy) **Blinding:** Open-label / Unblinded **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N\$: 500 participants (An interim analysis included 353 patients) **Number of Sites:** 48 institutions across Japan **Estimated Study Start:** 27-Nov-2020

(Start of retrospective data inclusion period) **Estimated Primary Completion:** 10-Jun-2022 (Data cutoff for interim analysis)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients with a diagnosis of untreated stage IV or recurrent non-small cell lung cancer (NSCLC). 2. Initiated first-line treatment with nivolumab plus ipilimumab with or without chemotherapy between November 27, 2020, and December 31, 2021. 3. Age $\geq$ 20 years.

4.2 Exclusion Criteria 1. In participants with non-squamous histology, those confirmed to be positive for EGFR gene mutation or ALK fusion gene (for which targeted tyrosine kinase inhibitors are indicated). 2. Participants who received any prior antineoplastic treatment as a first-line therapy for advanced or recurrent NSCLC prior to the initiation of the study regimen.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nivolumab plus Ipilimumab with or without platinum-based doublet chemotherapy. The exact treatment regimen, dosage, and schedule were independently determined at the treating physicians' discretion. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or discontinuation determined by the physician.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures routinely used in clinical practice. **Prohibited:** EGFR or ALK tyrosine kinase inhibitors (TKIs) as first-line therapy due to lack of target mutations.

6. Study Timeline & Visit Schedule

(Note: As this is a retrospective observational chart review, the visit schedule reflects standard clinical practice rather than a protocol-mandated cadence.) | Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Retrospective | Medical chart review, diagnosis verification, and mutational status (EGFR/ALK) check | | **Baseline** | Day 0 | Initiation of first-line nivolumab plus ipilimumab (with/without chemotherapy) | | **Interim** | Real-world clinical intervals | Assessment of tumor response, recording of AEs/irAEs | | **End of Study** | Follow-up | Assessment of Progression-Free Survival (PFS), date of second-line treatment, or death |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time from the date of initiation of nivolumab plus ipilimumab (with or without chemotherapy) to the last date of treatment, the date of second-line treatment initiation, or the date of death from any cause (used as proxies for PFS/OS in the real-world setting). **Measurement Method:** Retrospective electronic medical record (EMR) data extraction.

7.2 Secondary & Exploratory Endpoints * The rate of eligible participants who continued the treatment regimen during the observation period. * Incidence and severity of immune-related adverse events (irAEs) and treatment-related deaths.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective data extraction of safety profiles from medical records from the 48 participating institutions. **Serious Adverse Events (SAEs):** Specific tracking of grade 3-4 immune-related adverse events (which showed the highest incidence in the first month of therapy) and treatment-related mortality. **Data Monitoring Committee (DMC):** N/A (Observational, non-interventional design).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Patients who met all eligibility criteria and initiated treatment, generally requiring a minimum follow-up of 90 days (though data from patients who died or were lost to follow-up sooner were also included). **Sample Size Justification:** A target of 500 patients was selected based on feasibility to investigate actual clinical practice trends across multiple centers in Japan. **Handling of Missing Data:** Handled via standard epidemiological methodologies for retrospective real-world data; participants lost to follow-up were censored at the date of last contact.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: The study was conducted in accordance with the ethical principles of the Declaration of Helsinki and approved by the ethics committees at each of the 48 participating hospitals. **Monitoring Plan:** Retrospective data validation against source medical records. **Audit Trail:** Traceable electronic case report forms (eCRF) utilizing real-world demographic, staging, and treatment administration logs.

11. Study Identifiers & Governance

Primary ID: CA209-64A **Registry Number:** NCT05161325 **Sponsor/Lead Organization:** Bristol-Myers Squibb / Ono Pharmaceutical Co., Ltd. **Study Title:** LIGHT-NING **Official Regulatory Title:** An Observational Study of Effectiveness and Safety of the First-Line Nivolumab Plus Ipilimumab With or Without Chemotherapy for Advanced/Recurrent Non-Small Cell Lung Cancer in Japan

12. Clinical Framework (NSCLC Specific)

Primary Condition: Advanced/Recurrent Non-Small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Stage IV or recurrent NSCLC, EGFR/ALK negative **Medical Methodology:** First-Line Combination Immunotherapy **Optimization Target:** Disease control, Progression-Free Survival (PFS), and Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard first-line systemic anti-cancer therapy for NSCLC **Education Protocol (HCP):** N/A (Retrospective observation of standard clinical practice) **Education Protocol (Patient):** N/A (Routine clinical oncology counseling) **Quality Audit Mechanism:** Centralized medical record abstraction and data curation

14. Observation & Follow-Up Schedule

Total Duration: Median follow-up of 7.1 months (interquartile range, 5.0-9.7 months) in interim analysis **Onsite Visit Cadence:** Determined by the treating physician's standard real-world practice **Remote Monitoring Frequency:** N/A (Retrospective medical chart extraction) **Checklist Review Interval:** Continuous retrospective assessment until data cutoff

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				